sambe CONSULTING

CELEBRATING 20 YEARS

DECISION INTELLIGENCE, REPORTING & DATA PLATFORMS

SALES DATA PROCESSING AND BUSINESS INSIGHTS SYSTEM

Date: April 2026

Jabulane Prince Poulo

Objectives

The project aims to develop a complete data processing and reporting workflow that transforms raw transactional sales data into actionable business intelligence. Graduates must demonstrate proficiency across three core domains: Linux automation, Python data processing, and Excel analytics.

Executive Summary

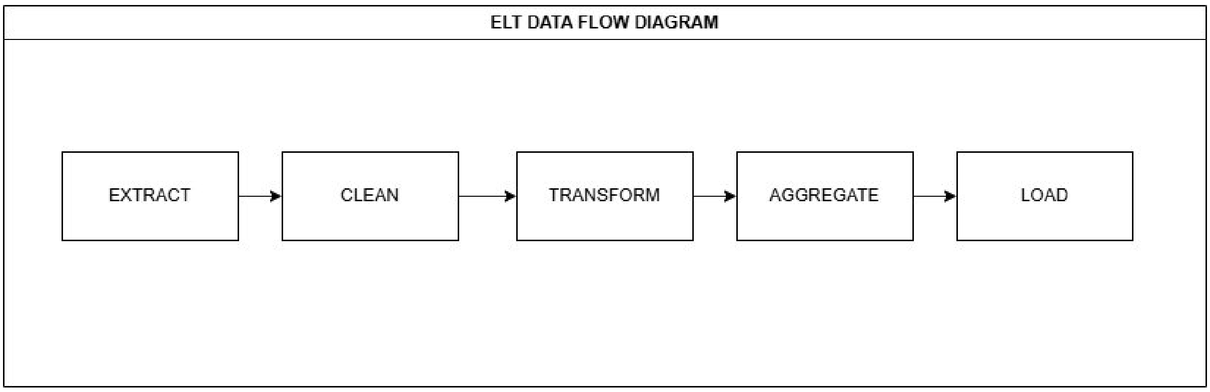
This report presents the findings from the Sales Data Processing and Business Insights System, a complete ETL (Extract, Transform, Load) pipeline developed to transform raw transactional sales data into actionable business intelligence. The system processes sales data across four regions (North, East, South, West), five product categories, and multiple salespeople to answer key business questions regarding regional performance, product revenue, sales trends, and individual performance.

Key findings

Metric	**Top Performer**	**Key findings**
Best Region	East	Has the highest revenue across all categories
Top Product	Phone	Highest revenue generator in Electronics
Best Performing Month	April	Best revenue generated
Top Salesperson	David	Leading performer in the East region

Data Processing Methodology

Pipeline Architecture



Data Cleaning

Data Quality Issue	Resolution Method
Missing quantity values	Found the median and filled the missing values with it
Missing prices values	Filled with category avarage
Missing Salesperson names	Labelled as Unknown
Invalid date formats	Standardised to datetime
Duplicate transactions	Removed completely

Calculated Fields

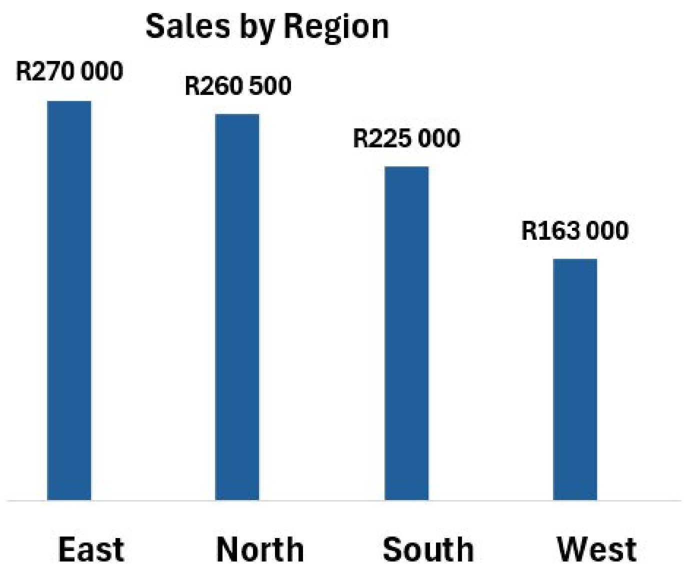
New Field	Formula	Business Value
Revenue	Quantity X Price	Enables all revenue-based analysis
Month	Extracted from date	Enables trend analysis

Business Analysis Findings

Regional Performance Analysis

Finding: The East region is the top-performing region.

Sales by Region:

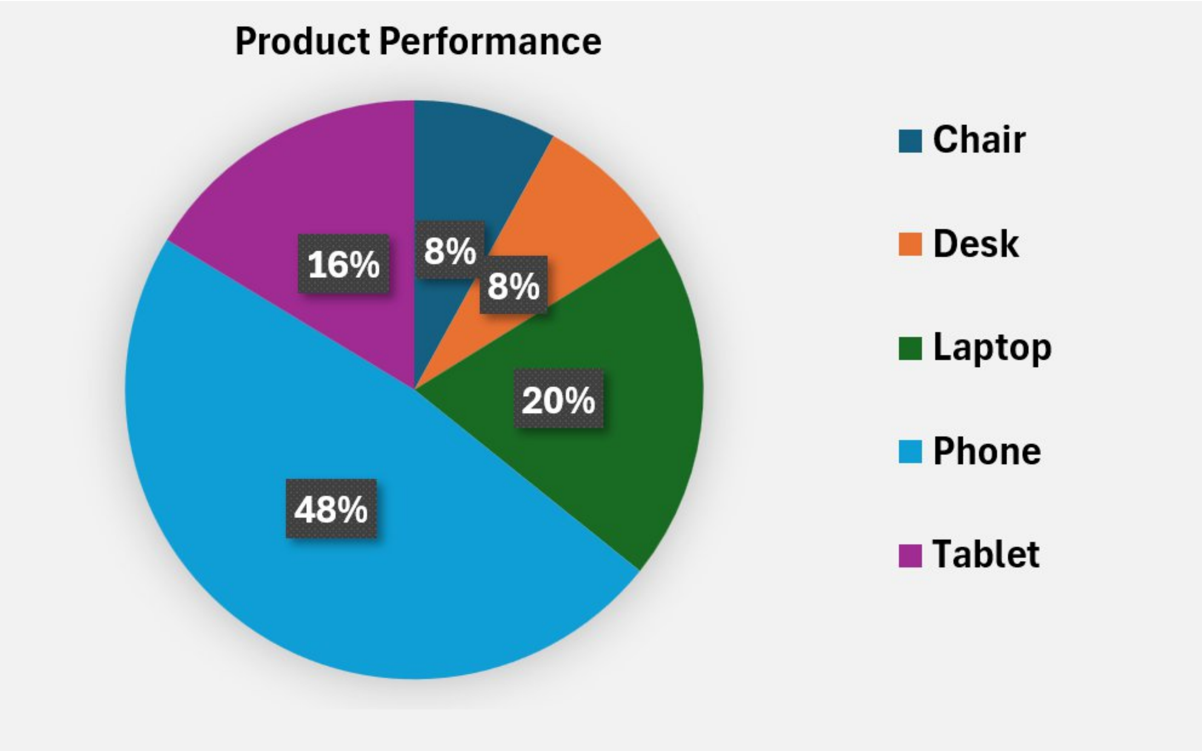


Analysis: The East region consistently outperforms other regions across all product categories. This suggests either higher customer demand in this region, more effective sales strategies, or both.

Business Implication: The sales strategies and practices used in the East region should be documented and replicated across other regions, particularly West region which shows the lowest performance.

Product Performance Analysis

Finding: Electronics category dominates total revenue, with Phone being the single highest revenue-generating product.

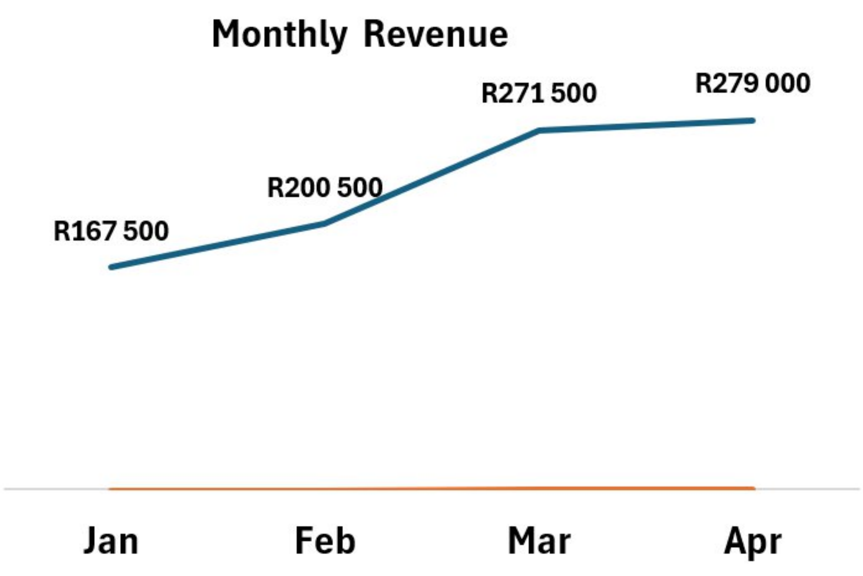


Analysis: Electronics products account for most of the total revenue. Within Electronics, the Phone product line is the strongest performer. Furniture sales, while present, contribute significantly less to overall revenue.

Business Implication: Inventory and marketing budgets should be allocated proportionally to revenue contribution, with Electronics receiving most resources.

Sales Trend Analysis

Finding: Sales show consistent growth from January through April, with March showing the strongest month-over-month increase.



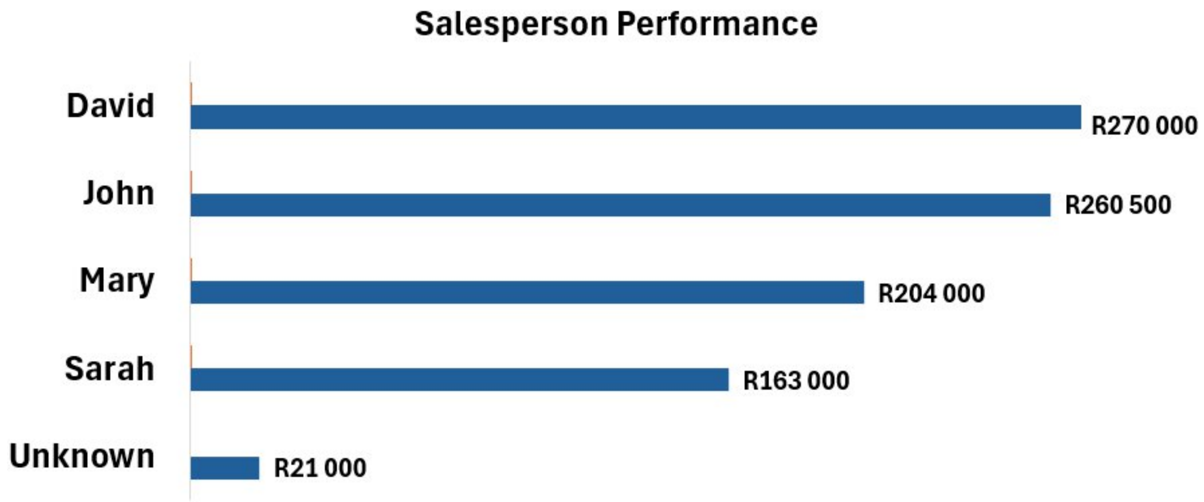
Analysis: The upward trajectory across all four months indicates growing demand or improved sales effectiveness. The strongest growth occurring in March suggests seasonal factors or successful promotional activities during that period.

Business Implication: The factors driving the March growth should be investigated and potentially replicated in other months. If seasonal, inventory should be increased ahead of future March periods.

Salesperson Performance Analysis

Finding: David is the top-performing salesperson, followed by John, Mary, and Sarah.

Salesperson Rankings:



Analysis: David, operating in the East region, leads all salespeople. Notably, the "Unknown" salesperson appears in the dataset, indicating data entry gaps where transactions were recorded without an associated salesperson.

Business Implication: The top performers should be recognised and their techniques shared with lower-performing salespeople. The "Unknown" records represent a process failure that needs to be addressed at the point of sale.

Actionable Business Recommendations

Recommendation 1: Replicate East Region Success Formula

Rationale: The East region consistently outperforms other regions. Understanding and replicating their success factors is the fastest path to improving overall company performance.

Success Metric: West region revenue increase of 20% within 3 months of implementation

Recommendation 2: Optimise Product Strategy

Rationale: Electronics (specifically Phones) drives most of the revenue. Resources should be allocated proportionally to revenue contribution.

Success Metric: Electronics revenue increase of 15% next quarter clear decision on Furniture line continuation.

Recommendation 3: Fix Data Quality at Source

Rationale: The presence of "Unknown" salesperson records prevents accurate performance tracking and may hide revenue attribution issues.

Success Metric: Zero "Unknown" salesperson records in next month's data; 100% completion rate for all required fields.

Conclusion

This project successfully delivers a complete data processing and business intelligence system that transforms raw transactional data into actionable insights. The solution:

1. **Automates** the entire ETL pipeline using Linux shell scripting and Python
2. **Cleans and standardises** messy raw data into analysis-ready format
3. **Provides** clear answers to all management questions regarding regions, products, trends, and personnel
4. **Delivers** an interactive Excel dashboard for ongoing self-service analytics
5. **Includes** four actionable recommendations backed by data evidence

The system is production-ready, fully documented, and can be scheduled to run automatically, ensuring management always has access to current sales insights without manual intervention.

Technology Stack

Component	Technology	Version
Operating System	Ubuntu (WSL)	22.04
Data Processing	Python	3.10+
Data Analysis Library	pandas	2.1.0
Automation	Bash	5.1+
Reporting	Microsoft Excel	365
Version Control	Git	2.40+

Glossary

Term	Definition
ETL	Extract, Transform, Load - Data pipeline process
venv	Virtual environment - isolated Python environment